BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** Code: D

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and over writing will not be awarded any marks.

Qi. The unit of coefficient of friction is …

(A) Kg
(B) N
(C) μ
(D) It has No units.

Qii. The average speed of a bus is $20ms^{-1}$, how far can it travel in 10 seconds.

(A) 100m
(B) 150m
(C) 200m
(D) 250m.

Qiii. The value of 'g' at the surface of the Moon is …

(A) $1.63ms^{-2}$
(B) $4.9ms^{-2}$
(C) $8.9ms^{-2}$
(D) $9.8ms^{-2}$

Qiv. The instrument best measures the internal diameter of a pipe is …

(A) Vernier caliper
(B) Meter rule
(C) Measuring tape
(D) Screw gauge

Qv. Which prefix represents a largest value?

(A) Giga
(B) Mega
(C) Exa
(D) Peta

Qvi. The area under a speed time graph represent…

(A) Acceleration
(B) Distance
(C) Velocity
(D) Change in acceleration

Qvii. When water at 0°C is heated, it contracts till temperature reaches.....

(A) 0°C
(B) 4°C
(C) 100°C
(D) 100k

Qviii. Which of the following has the highest thermal conductivity?.

(A) Wood
(B) Wool
(C) Air
(D) Water

Qix. The S.I unit of Latent heat is .....

(A) JK^{-1}
(B) JKg^{-1}
(C) $J\ K\ Kg^{-1}$
(D) J K kg,

Qx. A girl pushes to open a door perpendicularly with a force of 25N at 0.6m from the hinger, the torque is.....

(A) 15Nm
(B) 41.6Nm
(C) 25.6Nm
(D) 0 Nm

Qxi. Work done will be zero when the angle between force and displacement is ...

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Qxii. Young modulus is measured in units of ....

(A) Pa
(B) N
(C) Kgm^{-3}
(D) None of these

Section "B"

Time Allowed: 2 Hours 40 Minutes. | **Maximum Marks:** 32

QQ.2. Attempt any EIGHT parts. All parts carry equal marks.

i Write the following numbers in power of ten. (a) 7 nanometer (b). 96 megawatt (c). 43 picofarad (d). 2 mm

ii A ball is thrown upward with an initial speed of 5m/sec. What will be its speed when it returns to starting point.

iii Prove that Vf = vi + at

iv Why is the surface of a conveyor belt made rough?.

v How much energy is generated when mass of 1 g is completely converted into energy?

vi An object of mass 10kg is lifted vertically through a height of 5m at a constant speed. What is the gravitational potential energy gained by the object?.

vii Why a small needle sinks in water and huge ships travel easily in water without sinking?

viii Define work and its unit. Under what condition the work done will be maximum and minimum?.

ix State Hook's law and explain elastic limit

x Define the terms heat capacity, specific heat capacity and their units.

xi How woolen sweaters keep us warmer in winter?

Section "C"

Maximum Marks: 21

Note: Note: Attempt any THREE of the following questions. All questions carry equal marks.

QQ.3.

a. Explain Newton's second law and third law of motion with examples .

b. Two bodies of masses 3kg and 5kg are tied to a string which is passed over a pulley. If the pulley has no friction, find the acceleration of the bodies and tension in the string.

QQ.4.

a. What is stability?. Explain the states of equilibrium.

b. The mass of the Sun is 1.99×10^{30} kg and the radius of the Earth's orbit around the Sun is 1.5×10^{11} m. From this, calculate the orbital speed of the Earth?.

QQ.5.

a. State Pascal's principle. Prove that hydraulic lift is a force magnifier.

b. When a weight of 30N is hung from a wire of original length 2.0m, its new length becomes 2.20m. Calculate the force constant for the wire, if the elastic limit is not exceeded?.

QQ.6.

a. Explain how length of a substance increases due to rise in temperature ?.

b. What is the specific heat of a metal substance if 135kj of heat is needed to raise 4.1 kg of the metal from 18° C to 37.2°C?